When Overall AUC Hides Miscalibration - Evaluating Knowledge Tracing on Rarely Practiced Knowledge Components

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***Abstract*—Adaptive systems turn knowledge tracing (KT) probabilities into instructional decisions that need calibration, yet models are usually ranked by the area under the receiver operating characteristic curve (AUC). We present an evaluation protocol that groups knowledge components (KCs) by training frequency, reports each group's test support, measures calibration by expected calibration error (ECE) and Brier score decomposition, bounds the expected small-sample bias of ECE, and specifies eight leakage checks. On ASSISTments 2012, the final-checkpoint ECE of deep knowledge tracing (DKT) rose from 0.0602 on dense to 0.2333 on sparse KCs, by more than the bias bound, although sparse-group test support was limited. Restoring the validation-AUC checkpoint and scoring windows of 200 left a size-matched gap, but not one above the bias bound; for simpleKT, the rise (0.0203 to 0.0884) did not exceed the bound. At a fixed threshold, DKT's advances on sparse KCs were followed by incorrect responses more often than those on dense KCs in three of five runs. Junyi Academy had no sparse group. Its final-checkpoint ECE rose from 0.0389 on dense to 0.2513 on medium KCs (ΔECE* = 0.181). Under the validation-AUC checkpoint the four-partition ΔECE* was −0.0032, while fold 4 remained positive. On XES3G5M, the leakage audit found that a response's outcome entered the input for its other KC rows in training and scoring, so no model-based results are reported. KT evaluations informing instructional decisions should report calibration, sample support, and leakage checks by KC frequency.**

***Keywords*—knowledge tracing, calibration, knowledge components, adaptive learning, learning analytics, evaluation protocol**

I. INTRODUCTION

Adaptive practice platforms and intelligent tutoring systems use knowledge tracing (KT) to estimate how well a learner has mastered a skill and to decide whether to give more practice, assign remediation, or move on [1]–[3]. Such decisions often rest on the predicted probability p that the learner will answer the next item on the skill correctly: once p reaches a threshold τ, further practice on that skill is withheld. A rule of this kind takes p at face value. A prediction of 0.7 should mean that about seven in ten comparable responses are correct; in other words, the model must be well calibrated.

KT models, however, are usually compared by the area under the receiver operating characteristic curve (AUC) and accuracy computed over all test responses [4], [5]. AUC measures ranking, that is, whether correct responses tend to receive higher probabilities than incorrect ones; calibration is a separate property. The difference matters most for knowledge components (KCs) that learners rarely practice. In the training data used here, 18.9% of the KCs in ASSISTments 2012 and 21.5% of the XES3G5M KCs present in training had fewer than 100 training rows (Section IV.C), whereas frequently practiced KCs supplied most of the rows. Pooled metrics therefore mainly reflect frequently practiced KCs, and a model can rank responses well overall while assigning overconfident probabilities to rarely practiced KCs. Under a fixed threshold, such overconfidence appears as an advance decision followed by an incorrect next response; we call the proportion of these decisions the false-advance rate (FAR).

Earlier work has examined parts of this problem. Benchmark studies have standardized KT evaluation and documented label leakage [5], [6], and calibration, uncertainty, cold start, and mastery thresholds have been studied in their own right (Section II). This study combines these elements into one evaluation protocol for standard KT models; it does not propose a new model.

Applying the protocol to three public datasets, the study addresses three research questions:

RQ1: How does calibration vary across training-frequency groups and datasets?

RQ2: Under what conditions can a sparse-versus-dense calibration contrast be estimated?

RQ3: On ASSISTments 2012, does the direction of that contrast hold under alternative frequency cut points?

For RQ1, on ASSISTments 2012 the final-checkpoint expected calibration error (ECE) of deep knowledge tracing (DKT) was higher on sparse than on dense KCs by more than a bound on expected small-sample bias; a local Transformer model (T-KT) showed a smaller margin, and the difference for simpleKT [4] stayed within the bound. On Junyi Academy, DKT's final-checkpoint ECE was higher on medium than on dense KCs. Section IV reports the validation-AUC checkpoint separately. For RQ2, the contrast could be estimated only where the KC definition left a low-frequency tail with enough test responses: on ASSISTments 2012 and XES3G5M, but not on Junyi Academy, whose exercise-level KCs all had at least 100 training responses under the learner-based split. On XES3G5M, a failed leakage check then withheld all model-based results. For RQ3, the sign of the final-checkpoint contrast held for DKT and T-KT under two alternative cut grids, also after allowing for small-sample bias.

The paper makes three contributions. First, it specifies an

evaluation protocol with formal definitions. The protocol assigns KCs to training-frequency groups, including a strict cold-start group, labels each group by its test-sample support, states when a sparse contrast is estimable, and bounds the expected small-sample bias of group-level ECE. It also defines eight leakage checks: the archived audit applies the first seven to every dataset and partition, and applying the eighth to XES3G5M exposed label leakage in our own pipeline. Second, it applies the protocol to three public datasets and reports calibration and threshold errors by group, together with the cases in which the protocol withholds a conclusion. Third, it is supported by an analysis artifact that rebuilds selected diagnostic tables from saved summaries.

II. LITERATURE REVIEW

A. Knowledge tracing models and benchmarks

KT models a learner's changing proficiency from past interactions in order to predict the next response [3]. Bayesian knowledge tracing (BKT) tracks a latent mastery state for each skill [1]. Item response theory (IRT), in its one-parameter Rasch form [7], relates the probability of a correct answer to learner ability and item difficulty. DKT replaced hand-specified learning dynamics with a recurrent network and established next-response prediction as the standard task [2]. Attention-based successors, such as context-aware attentive knowledge tracing [8] and simpleKT [4], keep that task but replace recurrence with self-attention. The best model depends on the data: in a comparison on nine real-world datasets, logistic regression with a suitable feature set led on datasets of moderate size, whereas DKT led on the largest ones [9]. Uncertainty-aware models such as UKT represent knowledge states as distributions [10].

B. Evaluation rigor and leakage

Benchmark libraries such as pyKT have standardized preprocessing and model comparison and have shown that evaluation choices can distort reported discrimination [5]. One such choice concerns questions tagged with several KCs. When such a question is expanded into one row per KC and each row is scored separately, a model can use the answer to the same question from the previous row. In pyKT's experiments, this raised AUC by 8.38 and 13.09 percentage points on average on ASSISTments 2009 and Algebra 2005, respectively [5]. Badran and Preisach analyze the same form of label leakage [6], and our protocol defines this route as check L8. The protocol applies the same discipline to frequency groups and calibration: each KC's group and descriptors are computed from training data only, and no cut point or threshold is tuned on test data. Graph-based KT [11] and contrastive learning for KT [12] form related research lines that focus on new model architectures.

C. Four meanings of sparsity

The word "sparse" has at least four meanings in KT research. (a) Sparse attention: sparseKT applies k-sparse attention so that only a subset of past interactions receives nonzero weight [13]. (b) Sparse KC frequency: a KC with few observations in the training data. (c) New-student cold start: prediction for learners with little or no history at test time [14]. (d) Concept-level cold start: a KC that never appears in the training data, whether or not the learner is new. These problems are distinct: sparse attention can be applied to a frequently practiced skill, and a learner seen in training can meet a KC absent from it. This study addresses meanings (b) and (d) through training-frequency groups that include a strict cold-start group ($f_{\text{train}} = 0$).

D. Calibration and educational decisions

When predicted probabilities drive educational decisions, ranking alone is an insufficient evaluation [15]. Calibration asks whether a predicted probability matches the observed frequency of correct responses. ECE summarizes the mismatch over probability bins [16], and modern neural networks can be poorly calibrated even when they discriminate well [17]. The Brier score [18], Murphy's decomposition into uncertainty, resolution, and reliability [19], and the calibration–refinement view [20] separate miscalibration from the difficulty of the task, while reliability diagrams display the same comparison graphically.

Within KT, post-hoc per-item bias correction has been used to recover discrimination in a trained model [21], and selective prediction defers the least certain predictions to a teacher [22]. Mastery-learning research has examined false positives at the mastery threshold [23] and predictive performance when a student starts a skill [24]. Our threshold analysis complements selective prediction: it does not abstain, but measures how often the advances made under a fixed threshold are followed by an incorrect response, separately for each frequency group.

III. MATERIALS AND METHODS

Fig. 1 outlines the evaluation workflow; the leakage checks attached to its steps are defined in Section III.F.

A. Datasets

We used three de-identified public logs mapped to a common interaction schema: ASSISTments 2012 [25], Junyi Academy [26], and XES3G5M [27]. Preprocessing removed rows with missing learner, KC, or label identifiers, binarized correctness, parsed timestamps, and kept learners with at least two interactions. Table 1 reports the counts after these filters, which therefore differ from those of the raw public files. The processed logs contain 2,657,490 (ASSISTments 2012), 16,215,567 (Junyi Academy), and 6,413,353 (XES3G5M) rows.

Table 1. Cohort statistics after preprocessing (learner-based split; test events from fold 0). For XES3G5M, interactions and test events are KC-level rows (Section III.A).

Dataset	Learners	KCs	Interactions	Test events
ASSISTments 2012	27,806	265	2.66M	534,150
Junyi Academy	71,014	1,326	16.2M	3,269,022
XES3G5M	18,066	865	6.41M	1,282,422

ASSISTments 2012 is the 2012–2013 public release with affect [25], which is also supported by the pyKT toolkit [28]; each retained row carries one skill_id. Junyi Academy is the Kaggle Online Learning Activity dataset [26], which differs from the 2015 Junyi release. We used its ucid field, the content (exercise) identifier that the practice log assigns to

each item, as the KC; it is an operational identifier rather than a curated skill tag. Practice volume per exercise is high: across the whole log, the 1,326 exercises received about 12,200 responses each on average. In learner-based fold 0, every exercise had at least 100 training responses (Fig. 2), and the sparse group was empty in all four learner-based partitions, so Junyi Academy has no sparse group under the learner-based split.

XES3G5M was loaded from the authors' KC-level release: the training/validation file and the test file were concatenated and flattened to one row per valid sequence position, with padding, invalid labels, and rows with a negative KC identifier removed. The dataset paper reports 18,066 students, 7,652 questions, 865 KCs, and 5,549,635 question-level interactions [27]. Our file has 6,413,353 rows because a question tagged with several KCs is expanded into one row per KC; the number of distinct items remains 7,652, and all XES3G5M counts, including the 1,282,422 test events of fold 0 (Table 1), are KC-level rows. If the model input for one row contains the outcome recorded at another row of the same response, the target label is visible to the model [5], [6]. Our audit of this route (check L8, Section III.F) in the code used for the XES3G5M runs found that the KC rows of a response stay adjacent and that both training and scoring condition each prediction on the KC–response tokens of earlier rows. For the second and later KC rows of a multi-KC response (863,807 rows by the release's repeat flags, 13.5% of the file), the input therefore contains the outcome being predicted, in training and at test time, so L8 fails. Because removing this route requires retraining, which this study did not do, we report no model-based results for XES3G5M; the dataset still contributes the frequency distribution and estimability conditions (Fig. 2 and Section IV.C), which do not depend on predicted values. In ASSISTments 2012 and Junyi Academy, each released row records one response with one KC identifier and no expansion step was applied, so the route does not arise.

B. Splits and partitions

The learner-based split is the primary design. For each partition, learners were shuffled with the partition seed, and 20% were held out for testing, 10% for validation, and the rest for training, so the three sets share no learners. A temporal split served as a complementary check: all interactions were sorted by timestamp, and the earliest 70%, the next 10%, and the latest 20% formed the training, validation, and test sets. It was run once (seed 42), and this paper reports from it only the temporal distributions in Fig. 2; all performance results come from the learner-based split. In both designs, KC frequencies, and hence group membership, were computed from the training set only.

Deep models were trained with seeds 42, 2024, 2025, 2026, and 2027 (folds 0–4). On every dataset, seeds 2025 and 2026 share one learner partition (fold 2 = fold 3), so the five runs cover four distinct partitions, and the two runs on the shared partition were averaged before averaging across partitions. The Junyi Academy runs of the Transformer model (T-KT, Section III.C) were aligned differently: seeds 42 and 2024 were two initializations on fold 0, averaged first, and seeds 2025, 2026, and 2027 were run on folds 1, 2, and 4, so these labels are initialization seeds rather than split seeds. Unless stated otherwise, every mean ± standard deviation reported here is therefore an equal-weight summary over four learner-based partitions.

C. Models

Table 2 summarizes the three local models, which come from our own code. The first is a one-parameter IRT model, $p = \sigma(\theta - \beta)$, fitted on the training data [7]. Under the learner-based split, every test learner is unseen, and for unseen learners our implementation returns the global training base rate instead of using the KC difficulties β. It therefore predicts the same probability for every response, and its AUC is 0.5000 by construction. We therefore treat it as a base-rate reference rather than a ranking competitor and label it "Base rate" in the tables; this outcome reflects our implementation rather than IRT in general. The second model is DKT [2], a long short-term memory (LSTM) network over KC–response tokens. The third, T-KT, is a two-layer Transformer encoder over the same tokens without positional encoding. In the archived code, its self-attention has neither a causal nor a padding mask, so in training the context of each target includes that response and later ones in the window. Test predictions use only earlier responses, so none sees its own outcome or a later one; the flaw is confined to training but may itself affect calibration, so T-KT serves only as a local reference model, distinct from the published simpleKT, and no conclusion rests on it alone. At test time, DKT and T-KT condition each prediction on all of the learner's earlier test responses, although training windows hold at most 200 (Table 2), and each learner's first test response receives $p = 0.5$. On ASSISTments 2012, we also trained simpleKT [4] in its pyKT implementation [5] with the same seeds and partition files and one fixed configuration listed in the artifact. Its training script keeps the epoch with the highest validation AUC (patience 8, at most 50 epochs) and scores test sequences in 200-response windows. Because simpleKT also uses item identifiers, its AUC is not ranked against the local models. Its run logs were not archived, but its scored files match those of the local models in the mean test count of the dense, medium, and sparse groups.

The three local models were trained for a fixed number of epochs and evaluated at the final checkpoint, so test data were never used for model selection. Pre-specified checks reported with the calibration results instead restore the validation-AUC checkpoint, with patience 8 and at most 50 epochs, and score non-overlapping windows of 200. The exact code version of the original training runs was not recovered; Table 2 lists the settings that could be recovered, with the context row read from the archived code, and no missing setting was imputed. BKT [1] was not scored because our pyBKT 1.4.1 runs produced degenerate predictions, which are documented in the artifact.

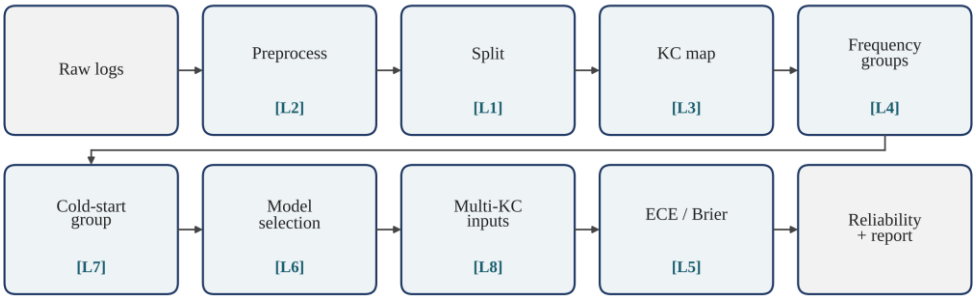


Fig. 1. Evaluation workflow. Bracketed tags give the leakage checks of Table 3 (L1–L7 numbered as in the archived audit records); they do not follow the left-to-right order. L8 applies only to datasets in which one response carries several KCs.

Table 2. Training settings recovered for the three local models (no missing setting was imputed; the context row is read from the archived code; simpleKT is described in Section III.C; SGD, stochastic gradient descent)

Setting	Base rate (1PL fit)	DKT	T-KT
Implementation	Local 1PL IRT	Local DKT	Local T-KT
Prediction input	None (global training base rate)	KC–response tokens	KC–response tokens
Architecture	1PL: sigmoid(θ−β); L2 0.01	LSTM; embedding 64, hidden 128	Transformer encoder; 2 layers, 4 heads, embedding 64
Input context (training; test)	—	Past, windows ≤ 200; all past	Window ≤ 200, unmasked; all past
Optimizer	SGD	Adam	Adam
Learning rate	0.01	1e−3	1e−3
Training length	10 epochs (fixed)	50 epochs (fixed)	50 epochs (fixed)
Batch size	512	64	64
Model selection	Final checkpoint	Final checkpoint	Final checkpoint

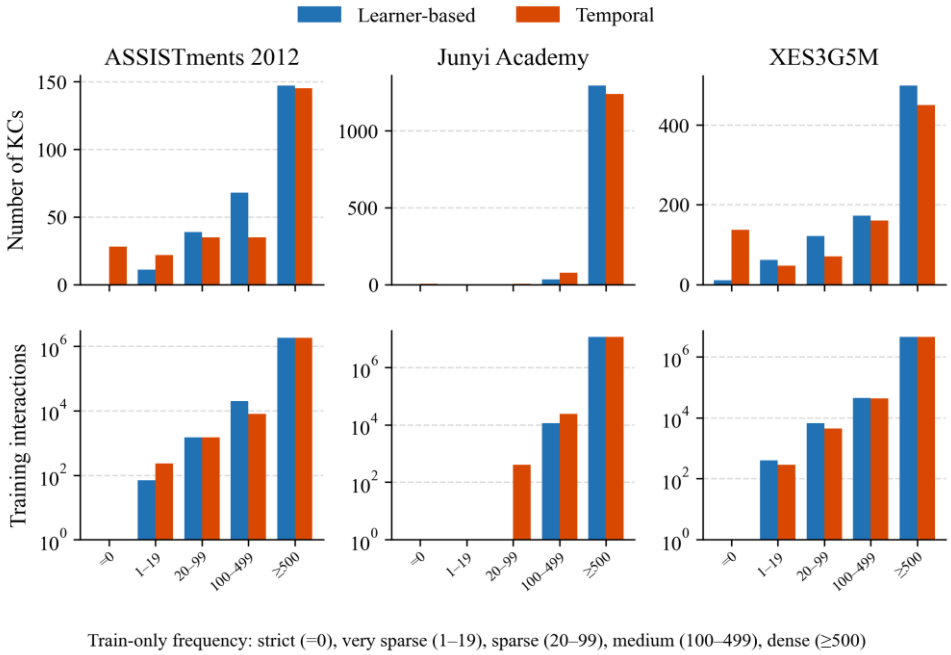


Fig. 2. Number of KCs (top) and number of training responses (bottom; sum of f_{train} in fold 0, logarithmic axis) in each training-frequency group under the learner-based and temporal splits.

D. Notation, frequency groups, and sample support

Each row of a processed log is a tuple (u, q, c, y). Learner u answered item q, c is the KC attached to the row, and $y \in \{0, 1\}$ records whether the response was correct. A partition divides the learners into disjoint training, validation, and test sets, and D_{test} denotes the test rows with a valid exported prediction. The training frequency $f_{train}(c)$ of KC c is its number of training rows. A group map g assigns each KC to one of five groups: strict cold start ($f_{train} = 0$), very sparse ($0 < f_{train} < 20$), sparse ($20 \le f_{train} < 100$), medium ($100 \le f_{train} < 500$), or dense ($f_{train} \ge 500$). Fig. 2 shows how many KCs and training responses fall in each group. The test rows of group k are

$$T_k = \{i \in D_{test} : g(c_i) = k\}, \quad N_k = |T_k| \tag{1}$$

and every group-level metric is computed on T_k alone. N_k is counted in each partition; the result tables report its mean over the four partitions and assign support labels from this mean, whereas the leakage audit (Section III.F) labels each partition. Analyses that use another subset, such as the threshold analysis, state their own N. A group is labeled Reliable (R) if $N \ge 1{,}000$, Limited (L) if $100 \le N < 1{,}000$, and

Insufficient (I) if $N < 100$. These cut points are conventions of this study that flag sample support; they are neither confidence intervals nor hypothesis tests. A metric can be computed for any nonempty group but is interpreted only for Reliable or Limited groups; Insufficient groups are reported descriptively. Likewise, a sparse-versus-dense contrast is computable whenever both groups are nonempty, and we treat it as estimable if and only if (E1) some KCs have $f_{\text{train}}(c) < 100$ in the training split and (E2) the sparse and the dense group are both at least Limited; E2 presupposes E1. In each partition, the protocol thus (i) assigns groups from training data, (ii) labels their support, (iii) checks E1 and E2, (iv) reports discrimination and calibration by group, with the bias bound of Section III.E, and (v) may apply one fixed threshold to estimate decision errors (Section III.G).

E. Calibration metrics

Within a group, let $p_i \in [0, 1]$ be the predicted probability of a correct response and y_i the observed outcome. With $M = 15$ equal-width bins, ECE is

$$\text{ECE} = \Sigma_m \, (n_m / N) \, |acc_m - conf_m| \tag{2}$$

where N is the number of rows in the group, n_m the number in bin m, and acc_m and $conf_m$ the observed proportion correct and the mean predicted probability in that bin. The Brier score is

$$\text{Brier} = (1/N) \, \Sigma_i \, (p_i - y_i)^2 \tag{3}$$

and its binned Murphy decomposition [19] is

$$\text{Brier} \approx \text{UNC} - \text{RES} + \text{REL} \tag{4}$$

with $\text{UNC} = \bar{y}(1 - \bar{y})$, $\text{RES} = (1/N) \, \Sigma_m \, n_m (acc_m - \bar{y})^2$, and $\text{REL} = (1/N) \, \Sigma_m \, n_m (conf_m - acc_m)^2$, where $\bar{y}$ is the proportion correct in the group. Because the components are computed on binned probabilities, they need not sum exactly to the Brier score. Lower ECE and REL indicate better calibration, and neither replaces AUC. The main calibration contrast is

$$\Delta\text{ECE} = \text{ECE}_{\text{sparse}} - \text{ECE}_{\text{dense}} \tag{5}$$

where each term is the mean over the four learner-based partitions. Partition-level standard deviations describe variation across partitions and are not confidence intervals for ΔECE. Binned ECE also depends on the number of bins and is biased upward when bins contain few responses [29], [30], so a small sparse group can by itself make ΔECE positive. To bound this bias, suppose the outcomes in a group were independent draws with the predicted probabilities. Then $acc_m - conf_m$ has mean zero and variance at most $1/(4n_m)$, so $\text{E}|acc_m - conf_m| \le 1/(2n_m^{1/2})$, and the Cauchy–Schwarz inequality gives

$$\text{E}[\text{ECE}] \le \text{ECE}_0(N) = \tfrac{1}{2} \, (M/N)^{1/2} \tag{6}$$

For $M = 15$, ECE_0 is 0.095 at $N = 415$ and 0.003 at $N = 523{,}971$. By the same argument, the expected ECE of any model exceeds its value at the bins' expected accuracies by at most ECE_0, so the conservative contrast

$$\Delta\text{ECE}^* = \text{ECE}_{\text{sparse}} - \text{ECE}_0(N_{\text{sparse}}) - \text{ECE}_{\text{dense}} \tag{7}$$

is positive only if the sparse–dense difference exceeds an upper bound on the expected small-sample bias. We evaluate ECE_0 at the mean N per partition and interpret a contrast only if $\Delta\text{ECE}^* > 0$, and then only by its sign. The bound is a worst case for independent outcomes; dependence among a learner's responses could raise it. Neither quantity is a confidence interval, and neither accounts for the variation of an observed ECE around its expected value.

F. Leakage checks

The protocol defines eight leakage checks. L1–L7 follow the numbering of the archived per-partition audit records, which cover only these seven. L1: training, validation, and test learners are disjoint. L2: every fitted transform uses training rows only; the row filtering of Section III.A fits no parameters. L3: the KC map is the static mapping supplied by the platform. L4: groups are assigned by applying the fixed cut points to f_{train} only. L5: no calibration map is fitted on test data. L6: no checkpoint or setting is selected with test data. L7: the strict cold-start group is defined by $f_{\text{train}}(c) = 0$. L8: when one response is expanded into several KC rows, the outcome recorded at one row does not enter the model input used to train on or score another row of the same response.

Table 3 reports the outcomes for the four learner-based partitions: checks L1–L7 passed in every partition. The archived audit covers the three local models; simpleKT reads the same partition files and selects epochs on validation learners only (Section III.C), so L1, L2, and L6 also hold for it by construction. The KC map (L3) is the released skill_id for ASSISTments 2012, the released KC mapping for XES3G5M, and the operational ucid for Junyi Academy. Check L8 does not apply to ASSISTments 2012 and Junyi Academy, where each released row records one response with one KC and no row was expanded. For XES3G5M, it was not part of the archived audit, and our audit of the code found that it fails (Section III.A). Test data were used only to compute the reported metrics.

Table 3. Leakage checks over the four learner-based partitions (L1–L7 numbered as in the archived audit records of the local models). PASS means that the condition held in every partition; parentheses give the occupancy of the affected group (I, Insufficient). L8 was audited from the code (Section III.A).

Check	Condition	ASSISTments 2012	Junyi Academy	XES3G5M
L1 Split	Training, validation, and test learners disjoint	PASS	PASS	PASS
L2 Preprocessing	Fitted transforms use training rows only	PASS	PASS	PASS
L3 KC map	Static, platform-provided mapping	PASS	PASS	PASS
L4 Frequency groups	Cut points applied to training counts only	PASS	PASS (sparse group empty)	PASS
L5 Calibration	No calibration map fitted on test data	PASS	PASS	PASS
L6 Model selection	No checkpoint or setting selected with test data	PASS	PASS	PASS
L7 Cold-start group	$f_{train} = 0$, defined from training data	PASS (empty in 3 partitions, I in 1)	PASS (empty)	PASS (I)
L8 Multi-KC rows	Outcome of a response hidden from the input for its other KC rows	N/A (one row per response)	N/A (one row per response)	FAIL (visible in training and test)

G. Secondary analyses

Two secondary analyses complement the group-level results. The first concerns KC structure. For each KC, we computed four training-only descriptors. The difficulty proxy is 1 − mean training correctness, an observed rate and not an IRT parameter. Item support is the number of distinct items per KC, learner exposure the number of distinct training learners per KC, and the curriculum-position proxy the median normalized position of the KC in learners' sequences. We report Spearman correlations of $\log(1 + f_{train})$ with the difficulty proxy (frequency–difficulty coupling) and with the curriculum-position proxy. A test-response-weighted regression relates T-KT's KC-level ECE to $\log(1 + f_{train})$ and the four descriptors, and a within-KC sparsification check reduces the training data of originally dense KCs.

The second is a simulated threshold analysis on the saved predictions. With one fixed threshold $\tau = 0.7$ for all groups, the simulated system advances the learner (skips remediation) when $p \geq \tau$ and assigns remediation otherwise. The false-advance rate is

$$\mathrm{FAR} = P(y = 0 \mid p \geq \tau) \tag{8}$$

that is, the proportion of advance decisions followed by an incorrect response; FAR equals one minus the precision of the advance decision, or its false discovery rate. We also report three related quantities: the false-positive rate FPR = $P(p \geq \tau \mid y = 0)$; the expected FAR implied by the model's own probabilities, $\mathrm{E}[1 - p \mid p \geq \tau]$; and the excess FAR, observed minus expected. The contrast $\Delta\mathrm{FAR} = \mathrm{FAR}_{sparse} - \mathrm{FAR}_{dense}$ is positive when advances on sparse KCs fail more often. FAR in (8) is computed over the N_{adv} responses with $p \geq \tau$, and FPR over the N_{inc} incorrect responses. Where prediction-level files exist, 95% intervals come from a KC-clustered percentile bootstrap (B = 2,000, with KCs resampled within the sparse and dense groups). Because y records the next response and not latent mastery, FAR is a simulated proxy for decision error.

IV. RESULTS

A. Overall discrimination

Table 4 reports overall AUC and accuracy (ACC). DKT had a higher AUC than T-KT on ASSISTments 2012 (0.6979 vs. 0.6837) and Junyi Academy (0.7320 vs. 0.7220). The simpleKT model, which also uses item identifiers, reached 0.7700 on the same ASSISTments partitions and is not ranked against the local models (Section III.C). These values are not directly comparable with pyKT benchmark results, whose preprocessing and split conventions differ from our learner-disjoint protocol [5]. Group-level AUC and accuracy are part of the artifact.

Table 4. Overall AUC and accuracy (ACC) on the learner-based split (mean ± sd over four partitions). The base-rate reference has an AUC of 0.5000 by construction (Section III.C); simpleKT, trained on ASSISTments 2012 only, is not ranked against the local models. XES3G5M is omitted because check L8 failed (Section III.A).

Dataset	Model	AUC	ACC
ASSISTments 2012	Base rate	0.5000	0.6973±0.0004
ASSISTments 2012	DKT	0.6979±0.0014	0.7182±0.0014
ASSISTments 2012	T-KT	0.6837±0.0025	0.6996±0.0032
ASSISTments 2012	simpleKT	0.7700±0.0013	0.7522±0.0014
Junyi Academy	Base rate	0.5000	0.7053±0.0018
Junyi Academy	DKT	0.7320±0.0009	0.7343±0.0015
Junyi Academy	T-KT	0.7220±0.0035	0.7267±0.0035

B. Calibration across frequency groups

On ASSISTments 2012, ECE rose from dense to sparse KCs for DKT, T-KT, and simpleKT (Table 5; RQ1), but by more than the bias bound only for the first two: from 0.0602 to 0.2333 for DKT ($\Delta\mathrm{ECE}^* = 0.078$) and from 0.1136 to 0.2280 for T-KT ($\Delta\mathrm{ECE}^* = 0.019$, about one partition SD). For simpleKT, it rose from 0.0203 to 0.0884, below $\mathrm{ECE}_0 = 0.095$ for the sparse group, so this contrast is not distinguishable from small-sample bias ($\Delta\mathrm{ECE}^* < 0$). Medium KCs lay in between for all three. The reliability term of the Brier decomposition rose in the same direction for DKT and T-KT (Table A1), for example from 0.0053 to 0.0624 for DKT. Those Table 5 figures use the final checkpoint. The original DKT weights were not recovered. A new final-epoch run, scored on the full history, gave fold-0 dense ECE 0.0562 and sparse ECE 0.2251. In the pre-specified DKT check, weights were restored from the validation-AUC checkpoint (patience 8, at most 50 epochs) and test rows were scored in non-overlapping windows of

200, dropping rows with an empty window. On fold 0 this gave sparse ECE 0.0916 and dense ECE 0.0070 (ΔECE = 0.0846; 430 sparse rows). Drawing dense learners without replacement until at least 430 dense events were available, then keeping exactly 430 events, over 2,000 draws, gave a median dense ECE of 0.0564 and a matched gap of 0.0351. A learner-cluster bootstrap on the same fold (B = 2,000, with 200 matched draws inside each replicate) gave a 95% interval of 0.0170 to 0.0821 for that gap. ΔECE* on this cell was −0.0088. The matched gap was positive on all four unique learner partitions, whereas ΔECE* was positive only for fold 1. Fold 2 and fold 3 are one partition and were not treated as two samples. The same validation checkpoint and window were used after T-KT was retrained with a causal mask and a padding mask. On fold 0 the size-matched gap was 0.0365 (95% interval 0.0233 to 0.0878; 430 sparse rows) and ΔECE* was −0.0102. The matched gap was positive on all four partitions (mean 0.0581), while mean ΔECE* was 0.0068 (SD 0.021). For simpleKT the same check left ΔECE* negative on every partition (four-partition mean −0.0379), and the fold-0 matched interval, −0.0049 to 0.0616, contains 0. The all-rows four-partition ECE of that rerun was 0.0251 on dense KCs and 0.0894 on sparse KCs, within 0.005 of the Table 5 pair.

On Junyi Academy, which has no sparse group, DKT's ECE rose from 0.0389 on dense to 0.2513 on medium KCs (3,706 medium test responses per partition, Reliable; ΔECE* = 0.181 with the medium group in place of the sparse one). Those Junyi DKT figures use the final checkpoint and the full test history. Under the validation-AUC checkpoint and windows of 200, four-partition mean ECE was 0.0345 on medium KCs and 0.0043 on dense KCs (ΔECE = 0.0302; ΔECE* = −0.0032; matched gap 0.0115). On fold 0 the matched gap was 0.0085, with a 95% interval of 0.0010 to 0.0434. For T-KT, at the final checkpoint and on the full test history, ECE rose from 0.0817 to 0.1136, a difference about equal to ECE_0 for this group (0.032). The causal-mask retraining was not run on Junyi Academy. Every medium KC had at least 100 training responses, so this contrast concerns moderately rather than rarely practiced KCs and is not counted as a sparse-group result.

The base-rate reference shows why ECE has to be read together with resolution. On dense ASSISTments KCs its ECE was only 0.0031, yet its resolution was zero (Table A1): a nearly constant predictor can appear well calibrated while carrying no ranking information. Because its predictions fall in one bin, its ECE equals the gap between that prediction and the observed proportion correct; on sparse KCs this gap (0.0777) exceeds the one-bin version of (6), 0.025, so it reflects a lower proportion correct rather than small-sample bias. The reference is thus neither a lower bound on attainable ECE nor a competitive model.

Fig. 3 shows the reliability diagrams for seed 42; the ASSISTments sparse curves rest on 444 responses over 15 bins and are noisy, so they serve only as visual diagnostics.

Table 5. Final-checkpoint, full-history ECE by training-frequency group (mean ± sd over four learner-based partitions; $M = 15$ equal-width bins). N is the mean number of test rows per partition after prediction export and group assignment, so totals differ from Table 1; ECE_0 is the bias bound (6) at that N. Flags: R, $N \geq 1{,}000$; L, $100 \leq N < 1{,}000$. The two T-KT values of 0.1136 (ASSISTments 2012, dense; Junyi Academy, medium) are separate results. XES3G5M is omitted because check L8 failed (Section III.A). The validation-AUC checks are reported in the text and are not in this table.

Dataset	Group	N	Flag	ECE_0	Base rate	DKT	T-KT	simpleKT
ASSISTments 2012	dense	523,971	R	0.0027	0.0031±0.0006	0.0602±0.0022	0.1136±0.0066	0.0203±0.0035
ASSISTments 2012	medium	5,963	R	0.0251	0.1598±0.0190	0.1621±0.0256	0.1541±0.0051	0.0262±0.0025
ASSISTments 2012	sparse	415	L	0.0951	0.0777±0.0555	0.2333±0.0084	0.2280±0.0197	0.0884±0.0187
Junyi Academy	dense	3,226,541	R	0.0011	0.0026±0.0018	0.0389±0.0022	0.0817±0.0070	—
Junyi Academy	medium	3,706	R	0.0318	0.1209±0.0082	0.2513±0.0217	0.1136±0.0155	—
Junyi Academy	sparse	—	empty	—	—	—	—	—

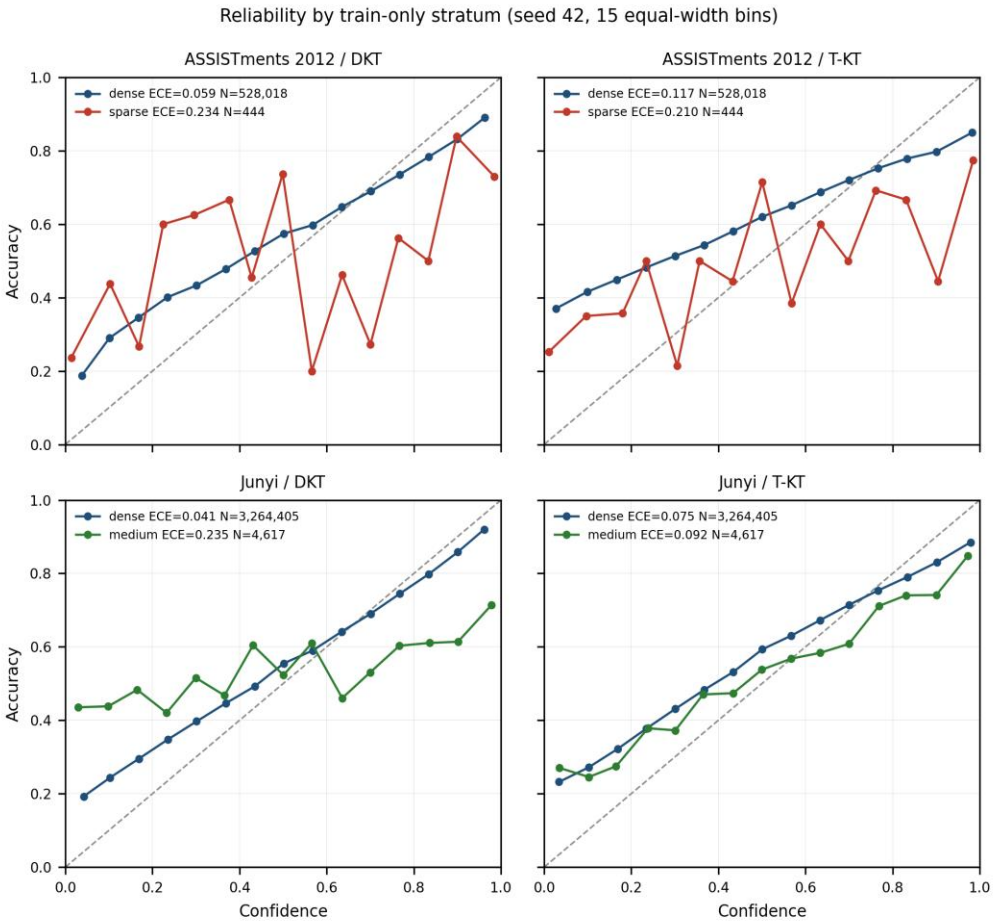


Fig. 3. Final-checkpoint reliability diagrams for DKT and T-KT (seed 42; full test history; 15 equal-width bins). Top: ASSISTments 2012, dense vs. sparse KCs. Bottom: Junyi Academy, dense vs. medium KCs (no sparse group). Tables 5 and A1 report the corresponding four-partition means. The validation-AUC checks are not plotted here.

C. Estimability and cold-start groups

Whether a sparse contrast can be estimated depends on the data (Table 6; RQ2). ASSISTments 2012 (18.9% of KCs below 100 training responses; 415 sparse test responses) and XES3G5M (21.5%; 1,969 KC-level rows) met conditions E1 and E2. Junyi Academy met neither, and the protocol reports an empty group instead of an estimate. These conditions depend on training and test counts, not on predicted values; for XES3G5M the counts are KC-level rows, and its contrast, although estimable by these counts, is not estimated because check L8 failed. The two smallest groups were seldom large enough to interpret. On ASSISTments 2012, the strict cold-start group had test responses in only one partition (4), and the very-sparse group averaged 17 per partition; both are Insufficient, so their ECE and AUC are descriptive only. On XES3G5M, the very-sparse group reached 114 test rows (Limited), and the strict cold-start group remained Insufficient.

The structural descriptors in Table 6 characterize the datasets, and none of them is treated as a necessary or sufficient condition for the direction of a contrast. Frequency and difficulty were negatively correlated on ASSISTments 2012 ($\rho = -0.227$, with sparse KCs harder) and Junyi Academy ($\rho = -0.416$) but weakly positively correlated on XES3G5M ($\rho = 0.110$).

Table 6. Conditions for estimating a sparse-versus-dense contrast, structural context, and calibration direction. E1 is computed on learner-based fold 0 over the KCs present in training; test-row counts are means over four learner-based partitions. Part B uses learner-based fold 0 (KCs with test rows; for XES3G5M, KCs present in training). Very sparse: $0 < f_{train} < 20$; strict cold start: $f_{train} = 0$. IQR, interquartile range. $\Delta ECE^* > 0$ refers to (7); in the medium rows, the medium group takes the place of the sparse one. XES3G5M model-based cells are not reported (check L8 failed).

Condition	ASSISTments 2012	Junyi Academy	XES3G5M
A. Estimability (counts only)			
Low-frequency tail: share of training KCs with $f_{train} < 100$ (E1)	18.9%	0%	21.5%
Sparse-group test rows (E2)	415 (L)	empty	1,969 (R)
Very-sparse group test rows	17 (I)	empty	114 (L)
Strict cold-start group test rows	4, one partition (I)	empty	I (N < 100)
B. Context (training data only)			
Frequency–difficulty correlation	$\rho = -0.227$ (sparse harder)	$\rho = -0.416$	$\rho = +0.110$ (weak, inverted)
Item support: median distinct items per KC	44.5 (dense 198, sparse 1)	18 (IQR 5)	3 (dense 9, sparse 1)
Curriculum-position correlation	$\rho = -0.308$	$\rho = -0.324$	$\rho = -0.126$
C. Calibration (model-based)			

Condition	ASSISTments 2012	Junyi Academy	XES3G5M
ECE, dense to sparse (DKT; T-KT)	rises; rises	no sparse group	not reported
ΔECE* > 0, sparse (DKT; T-KT)	yes; yes	—	not reported
ECE, dense to medium (DKT; T-KT)	rises; rises	rises; rises	not reported
ΔECE* > 0, medium (DKT; T-KT)	yes; yes	yes; ≈ 0	not reported

D. Robustness to frequency cut points

Under two alternative grids of training-frequency cut points, the final-checkpoint ASSISTments contrast ΔECE in (5) stayed positive for both models (Table A2). Its size, however, shrank as the sparse group grew: for T-KT it was 0.1260 with 303 sparse responses, 0.1145 with 415, and 0.1049 with 685, and for DKT it was 0.1853, 0.1730, and 0.1487. Two explanations fit this pattern and cannot be separated here. Binned ECE is biased upward when bins hold few responses [29], [30], and wider grids also admit KCs that are less sparse. ΔECE* in (7) stayed positive under every grid (0.074 to 0.078 for DKT; 0.015 to 0.031 for T-KT). Table A2 thus shows that, at the final checkpoint, the sign did not depend on the three grids examined; these grids were not repeated at the validation-AUC checkpoint. As a check on the grouping, the table is neither an interval estimate nor a test of the population contrast (RQ3).

E. KC-level associations and controlled sparsification

A test-response-weighted regression of T-KT's KC-level ECE on $\log(1 + f_{\text{train}})$ and the four descriptors compares different KCs in learner-based fold 0. The standardized frequency coefficient (95% interval, KC-clustered standard errors) was negative on ASSISTments 2012 (−0.063 [−0.081, −0.045]; 260 KCs) and Junyi Academy (−0.006 [−0.009, −0.003]; 1,326 KCs). Difficulty (positive) and curriculum position (negative) were also associated with ECE on both datasets, and item support and learner exposure on ASSISTments 2012. Because T-KT was trained and scored under different contexts (Section III.C), these associations describe that implementation only.

Reducing the training data of originally dense KCs did not produce a uniform direction. The check covered 12 cells (ASSISTments 2012 and Junyi Academy, DKT and T-KT, and reductions to 500, 100, and 50 training rows; 30 originally dense KCs each; seed 42). Three cells, all for T-KT on Junyi Academy, had 95% KC-bootstrap percentile intervals (B = 10,000) entirely above zero, four had intervals entirely below zero, and five included zero; no DKT cell had an interval entirely above zero. The per-cell grid is in the artifact. The group contrast, the regression, and this check estimate different quantities, so their signs need not agree.

F. Threshold analysis

At a fixed threshold of $\tau = 0.7$, advances on sparse ASSISTments KCs were followed by an incorrect response more often than advances on dense KCs. For DKT in the seed-42 partition, FAR was 0.200 on dense and 0.296 on sparse KCs (ΔFAR = 0.096, 95% confidence interval [0.019, 0.175]; Table A3): about 30 of every 100 advances on rarely practiced KCs were followed by an incorrect answer, against 20 of every 100 on frequently practiced KCs, whereas the model's own probabilities implied an error rate of 5.7% for those sparse advances. For T-KT, ΔFAR in that partition was 0.072 [0.006, 0.138]. Across the five runs, ΔFAR was positive in three for DKT and in all five for T-KT. Over the four distinct partitions (Table A4), the mean was 0.040 for DKT (range −0.011 to 0.096) and 0.056 for T-KT; DKT's two runs on the partition shared by seeds 2025 and 2026 differed in sign (−0.003 and 0.012). Because two DKT runs had a negative ΔFAR and both lower confidence bounds lie close to zero, we interpret only the direction of ΔFAR.

V. DISCUSSION

A. What the results show

Three patterns stand out. First, overall AUC did not reveal the calibration differences between frequency groups. On ASSISTments 2012, DKT's final-checkpoint ECE was higher on sparse than on dense KCs by more than the bias bound, whereas the validation-AUC checkpoint left a size-matched gap; its four-partition ΔECE* was not positive, while fold 1 remained positive; T-KT's final-checkpoint margin was smaller, and after retraining with a causal mask the fold-0 size-matched interval still excluded 0 while ΔECE* did not; simpleKT's contrast remained inside the bound, and its fold-0 matched interval contained 0; on Junyi Academy, DKT's final-checkpoint ECE was higher on medium than on dense KCs, whereas the validation-AUC checkpoint left a size-matched gap whose fold-0 interval excluded 0; its four-partition ΔECE* was not positive, while fold 4 remained positive. Second, at $\tau = 0.7$, advances on sparse ASSISTments KCs were followed by an incorrect response more often than advances on dense KCs, in three of five runs for DKT and in all five for T-KT, whereas in the seed-42 partition the models' own probabilities implied far fewer errors on sparse KCs (Table A3). Third, what can be concluded depends on the dataset and the model. Junyi Academy offered no sparse group, and on XES3G5M the leakage audit failed, so the protocol withholds a sparse-group conclusion for both, for different reasons. The XES3G5M case shows why check L8 belongs in such a protocol: our pipeline treated each KC row of a multi-KC response as a separate interaction, so a response's outcome entered the input for its other rows. Calibration on rarely practiced KCs is therefore something to check for each dataset and model rather than a property to assume.

B. Why the datasets differ

The datasets differed first in whether a sparse contrast could be estimated at all (Table 6). Junyi Academy's operational KC is the exercise (ucid), and practice per exercise was so high that, under the learner-based split, even the least practiced exercises had at least 100 training responses, which left the sparse group empty. ASSISTments 2012 and XES3G5M both met the estimability conditions but differ in frequency–difficulty coupling, item support, and curriculum position, which are reported as context; their

effect on the contrast cannot be judged until the XES3G5M models are retrained without leakage. The models differed as well: simpleKT, which differs from DKT in architecture, inputs, test-time windows, and checkpoint selection, showed no contrast beyond the bias bound, so the pattern seen for DKT need not hold for every KT model. The within-KC sparsification check did not reproduce the observational ASSISTments gradient for either local model, so the study identifies no mechanism behind it. Curriculum hierarchy, tagging granularity, ceiling effects, and item semantics remain untested candidate explanations.

C. Implications for adaptive learning platforms

For platforms that use KT probabilities to skip practice, assign remediation, or declare mastery, the results point to four checks before a single global threshold is trusted. (a) Occupancy: count KCs and test responses by training frequency in the split actually used, and report empty or low-support groups. (b) Calibration: report ECE and the Brier decomposition by group alongside AUC, and compare small-group ECE with the bias bound (6). (c) Decision error: if a fixed threshold is evaluated, report FAR and FPR together with their denominators. (d) Leakage and alignment: verify the checks in Table 3 in training as well as in scoring; when a response carries several KCs, construct the model input so that no KC row of the response reveals its outcome to another; in attention-based models, mask the target and later responses during training; and match each exported prediction to the right learner, item, KC, and sequence position.

Post-hoc calibration methods such as temperature scaling [17] and Platt scaling [31] were outside our scope; whether they close the gap on sparse KCs remains open.

D. Limitations

Several limitations qualify these findings. The outcome is the correctness of the next response, an imperfect indicator of mastery, and the threshold analysis replays logged data instead of testing real instructional decisions; no classroom trial was conducted. Only ASSISTments 2012 yields a sparse-versus-dense contrast from predictions that passed every applicable check, and its sparse group is Limited; Junyi Academy has no sparse group, and XES3G5M failed check L8. The protocol is thus applied to three datasets, but the contrast is not replicated. The only scored classical model, our 1PL IRT implementation, reduces to a global base rate under the learner-based split, and BKT was not scored. The code version of the local training runs was not recovered, and T-KT in Table 5 was trained without causal masking (Section III.C). The retraining reported above adds that mask together with the validation checkpoint and the 200-response window, so the change in ECE is not a mask-only effect. Table 5 scores DKT and T-KT with each learner's full earlier test history, which can exceed their 200-response training windows, and rarely practiced KCs tend to occupy later normalized positions in learners' sequences (Table 6). For DKT, scoring the same final-epoch weights in windows of 200 did not remove the positive raw ΔECE; the large drop in ECE came from restoring the validation-AUC checkpoint. The KC-level regression was run for T-KT only. The Table 5 simpleKT runs covered ASSISTments 2012 only and have no archived logs. The validation-checkpoint rerun reported above has a train log for each partition. Each dataset has four distinct learner-based partitions rather than five (Section III.B), and the temporal split was run once. The bias bound (6) assumes independent outcomes. For DKT on ASSISTments 2012, the validation-checkpoint check reported above includes a size-matched dense reference and a learner-cluster interval on fold 0. The same size-matched reference and fold-0 interval were later computed for the masked T-KT retraining, for simpleKT, and for Junyi DKT, as reported above. ΔECE* remains a reference bound, not a confidence interval. The XES3G5M models must be retrained with a corrected input construction before that dataset can contribute a calibration contrast. Junyi Academy was analyzed only at the exercise level. Graph-based, contrastive, and other self-supervised KT models were not evaluated. Finally, none of the analyses identifies a causal effect of training frequency on calibration.

VI. CONCLUSION

Overall AUC does not show how calibration changes for rarely practiced KCs. On ASSISTments 2012, DKT's final-checkpoint ECE was higher on sparse than on dense KCs by more than a bound on expected small-sample bias, although the sparse group had limited test support. Under the validation-AUC checkpoint the size-matched gap remained positive, the four-partition ΔECE* did not, and fold 1 remained positive. At a fixed threshold, its advances on sparse KCs were followed by an incorrect response more often than those on dense KCs in three of five runs; after the local Transformer was retrained with a causal mask, its fold-0 size-matched interval still excluded 0, while ΔECE* did not, and simpleKT stayed within the bound, with a fold-0 matched interval that contains 0. On Junyi Academy, which has no sparse group, DKT's final-checkpoint ECE rose from dense to medium KCs; under the validation-AUC checkpoint the size-matched gap remained positive, the four-partition ΔECE* did not, and fold 4 remained positive. On XES3G5M, the leakage audit withheld all model-based results. KT-based systems that turn probabilities into instructional decisions should report calibration and sample support by training frequency, with the bias bound and leakage checks, before relying on a single global threshold. The study contributes an evaluation protocol and diagnostic evidence, not a new KT model or a classroom intervention.

APPENDIX

Table A1. Final-checkpoint, full-history Brier score and Murphy decomposition by training-frequency group (mean ± sd over four learner-based partitions; $M = 15$). Brier ≈ UNC − RES + REL; the base-rate reference has RES = 0 by construction. XES3G5M is omitted because check L8 failed (Section III.A). The validation-AUC checks are not in this table.

Dataset	Model	Group	N	Flag	Brier	UNC	REL	RES
ASSISTments 2012	Base rate	dense	523,971	R	0.2103±0.0003	0.2103±0.0003	0.0000±0.0000	0.0000

Dataset	Model	Group	N	Flag	Brier	UNC	REL	RES
ASSISTments 2012	Base rate	medium	5,963	R	0.2742±0.0075	0.2484±0.0013	0.0258±0.0062	0.0000
ASSISTments 2012	Base rate	sparse	415	L	0.2420±0.0218	0.2337±0.0129	0.0084±0.0090	0.0000
ASSISTments 2012	DKT	dense	523,971	R	0.1931±0.0010	0.2103±0.0003	0.0053±0.0005	0.0223±0.0004
ASSISTments 2012	DKT	medium	5,963	R	0.2297±0.0144	0.2484±0.0013	0.0301±0.0086	0.0479±0.0053
ASSISTments 2012	DKT	sparse	415	L	0.2502±0.0098	0.2337±0.0129	0.0624±0.0038	0.0451±0.0037
ASSISTments 2012	T-KT	dense	523,971	R	0.2116±0.0019	0.2103±0.0003	0.0201±0.0015	0.0186±0.0004
ASSISTments 2012	T-KT	medium	5,963	R	0.2272±0.0086	0.2484±0.0013	0.0273±0.0021	0.0478±0.0053
ASSISTments 2012	T-KT	sparse	415	L	0.2525±0.0116	0.2337±0.0129	0.0594±0.0104	0.0392±0.0088
Junyi Academy	Base rate	dense	3,226,541	R	0.2078±0.0007	0.2078±0.0007	0.0000±0.0000	0.0000
Junyi Academy	Base rate	medium	3,706	R	0.2579±0.0033	0.2432±0.0013	0.0147±0.0021	0.0000
Junyi Academy	DKT	dense	3,226,541	R	0.1804±0.0008	0.2078±0.0007	0.0022±0.0002	0.0294±0.0002
Junyi Academy	DKT	medium	3,706	R	0.3077±0.0144	0.2432±0.0013	0.0741±0.0123	0.0091±0.0033
Junyi Academy	T-KT	dense	3,226,541	R	0.1892±0.0034	0.2078±0.0007	0.0091±0.0022	0.0275±0.0009
Junyi Academy	T-KT	medium	3,706	R	0.2347±0.0074	0.2432±0.0013	0.0163±0.0040	0.0251±0.0045

Table A2. Final-checkpoint, full-history dense-to-sparse ECE difference on ASSISTments 2012 under three training-frequency cut grids (four-partition means; dense: $f_{\text{train}} \geq t_2$; sparse: $t_0 \leq f_{\text{train}} < t_1$). The main grid matches Table 5. ΔECE uses unrounded values; ΔECE* in (7) uses the values shown, with ECE_0 at N sparse. simpleKT was not regrouped. These grids were not repeated at the validation-AUC checkpoint.

Model	Cut grid (t0/t1/t2)	Dense ECE	N dense	Sparse ECE	N sparse	ΔECE	ΔECE*
T-KT	Alt-1 (10/50/250)	0.1138	528,522	0.2397	303	0.1260	0.015
T-KT	Main (20/100/500) (Table 5)	0.1136	523,971	0.2280	415	0.1145	0.019
T-KT	Alt-2 (30/150/750)	0.1135	522,485	0.2184	685	0.1049	0.031
DKT	Alt-1 (10/50/250)	0.0610	528,522	0.2463	303	0.1853	0.074
DKT	Main (20/100/500) (Table 5)	0.0602	523,971	0.2333	415	0.1730	0.078
DKT	Alt-2 (30/150/750)	0.0600	522,485	0.2087	685	0.1487	0.075

Table A3. Threshold analysis at $\tau = 0.7$ on ASSISTments 2012 (seed 42, fold 0; sparse group Limited). FAR intervals: KC-clustered percentile bootstrap, B = 2,000. ΔFAR 95% CI: T-KT [0.006, 0.138]; DKT [0.019, 0.175].

Model	Group	N	N_{adv}	N_{inc}	FAR [95% CI]	E[FAR]	Excess	FPR
T-KT	dense	528,018	284,326	158,623	0.196 [0.186–0.208]	0.113	0.083	0.352
T-KT	sparse	444	235	197	0.268 [0.202–0.337]	0.050	0.218	0.320
DKT	dense	528,018	315,650	158,623	0.200 [0.190–0.211]	0.147	0.053	0.398
DKT	sparse	444	243	197	0.296 [0.221–0.383]	0.057	0.239	0.365

Table A4. ΔFAR at $\tau = 0.7$ on ASSISTments 2012 across the four learner-based partitions (sparse group). Seeds 2025 and 2026 share a partition and are averaged first; SD is over the four partition values. Five-run means: ΔFAR 0.047 (sd 0.033) for T-KT and 0.033 (sd 0.048) for DKT, with ΔFAR > 0 in 3 of 5 DKT runs; the sparse-group N (413), N_{adv}, and N_{inc} are also five-run means (four-partition N: 415, as in Table 5).

Model	Mean ΔFAR	SD	Range	Partitions with ΔFAR > 0	Mean N	Mean N_{adv}	Mean N_{inc}
T-KT	0.056	0.031	0.015 to 0.087	4/4	413	227	155
DKT	0.040	0.052	−0.011 to 0.096	3/4	413	226	155

Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

Author 1: Conceptualization, Methodology, Software, Formal analysis, and Writing (led diagnostic design, experiments, and manuscript). Authors 2 and 3: Software (data processing and baseline runs). Author 4: Methodology (methodological review). Author 5: Supervision and Writing (manuscript revision). All authors approved the final version.

Ethical Statement

This study is a secondary analysis of three publicly released educational datasets (ASSISTments 2012, Junyi Academy, and XES3G5M) that the providers de-identified before release. We did not recruit participants, contact learners, parents, guardians, or teachers, or conduct any classroom intervention, and we collected no new consent forms, logs, recordings, or personal data. No direct identifiers were accessed, and no attempt was made to re-identify any learner. The data were used under the providers' terms; Junyi Academy releases its data under CC BY-NC-SA 4.0, and XES3G5M is released under the MIT license. The study was neither commissioned by nor conducted with an educational technology company. Under the research policy of the authors' institution, a secondary analysis of publicly released, de-identified data does not require prospective human-subjects review, so no ethics protocol number applies. No exemption from data-protection law is claimed.

Data and Code Availability

The analysis artifact contains configuration files, preprocessing, training, and audit scripts, the saved summaries behind the reported tables, per-partition audit records, and scripts that, without retraining, rebuild selected diagnostic tables and recompute the bias bound (6) and the count-based rows of Table 6 from these files. It excludes the raw datasets, available from their providers, the prediction files, and the code of the controlled sparsification check, which was not archived; retraining requires those datasets and may not reproduce the reported values exactly. An anonymized copy is available to reviewers through the editor, and a public repository will be released upon acceptance.

Generative AI Statement

During manuscript preparation, the authors used Cursor 3.18.9 (with Grok 4.6 and Grok 4.7) for language polishing,

formatting, consistency checking, and preparation of reproducibility prompts; ChatGPT (OpenAI; GPT-5.6) to edit an intermediate revision and to compare revised versions; and Claude (Anthropic; Claude Opus 5.5) to review the manuscript and the analysis code and to draft structural and language revisions. No AI tool was used to generate or alter experimental results. The authors checked every AI-assisted passage and code finding against the underlying code and results, revised it, and take full responsibility for the content. No AI tool is listed as an author.

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